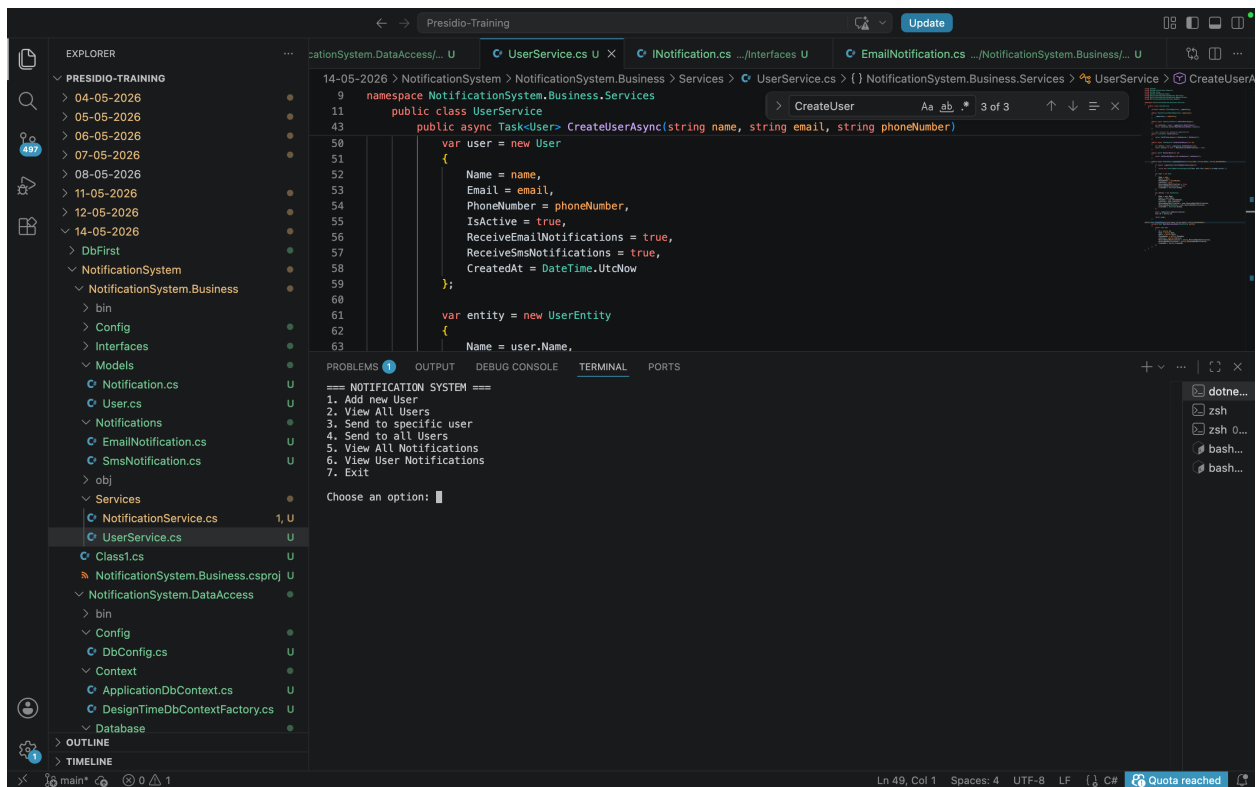


Notification System with EntityFrameworkCore

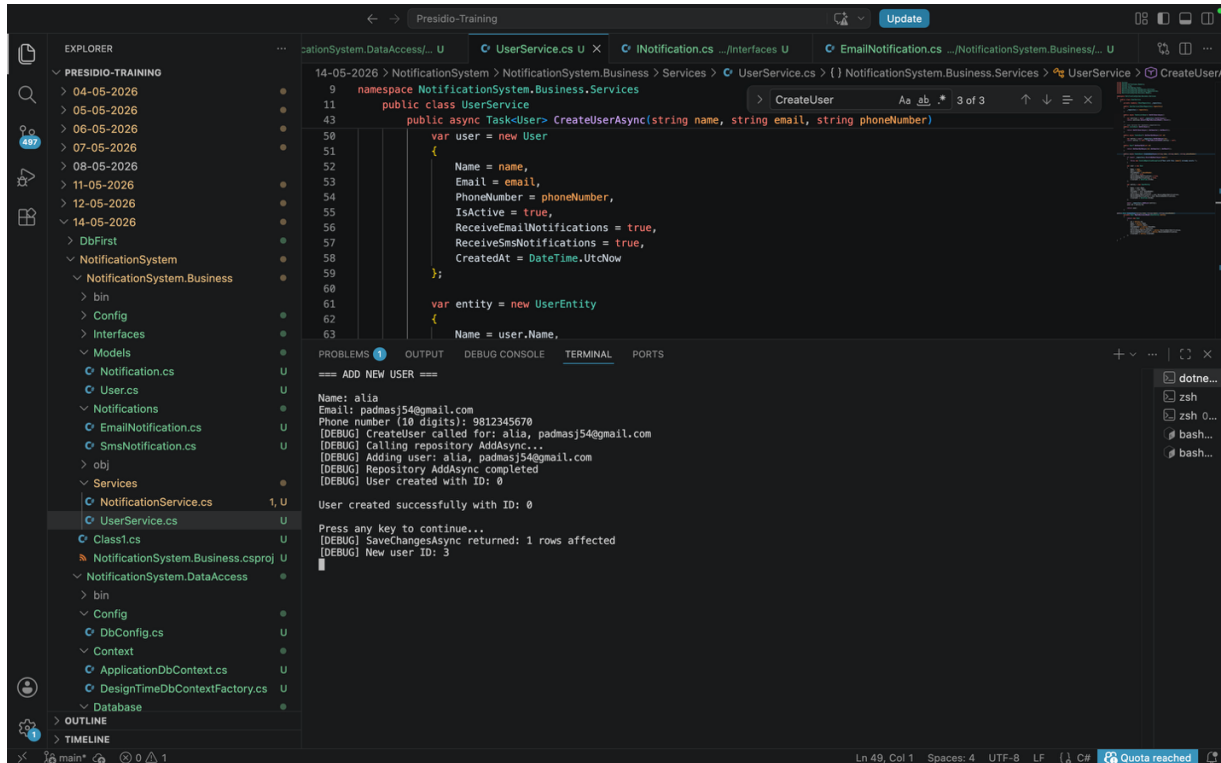
Source Code GitHub Url - <https://github.com/Padmavathi-SJ/Presidio-training/tree/main/14-05-2026/NotificationSystem>

OUTPUTS:

Menu:



Add New user:



```
14-05-2026 > NotificationSystem > NotificationSystem.Business > Services > UserService.cs > {} NotificationSystem.Business.Services > UserService > CreateUserAsync
9 namespace NotificationSystem.Business.Services
11 public class UserService
43 public async Task<User> CreateUserAsync(string name, string email, string phoneNumber)
50 {
51     var user = new User
52     {
53         Name = name,
54         Email = email,
55         PhoneNumber = phoneNumber,
56         IsActive = true,
57         ReceiveEmailNotifications = true,
58         ReceiveSmsNotifications = true,
59         CreatedAt = DateTime.UtcNow
60     };
61     var entity = new UserEntity
62     {
63         Name = user.Name,
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

=== ADD NEW USER ===

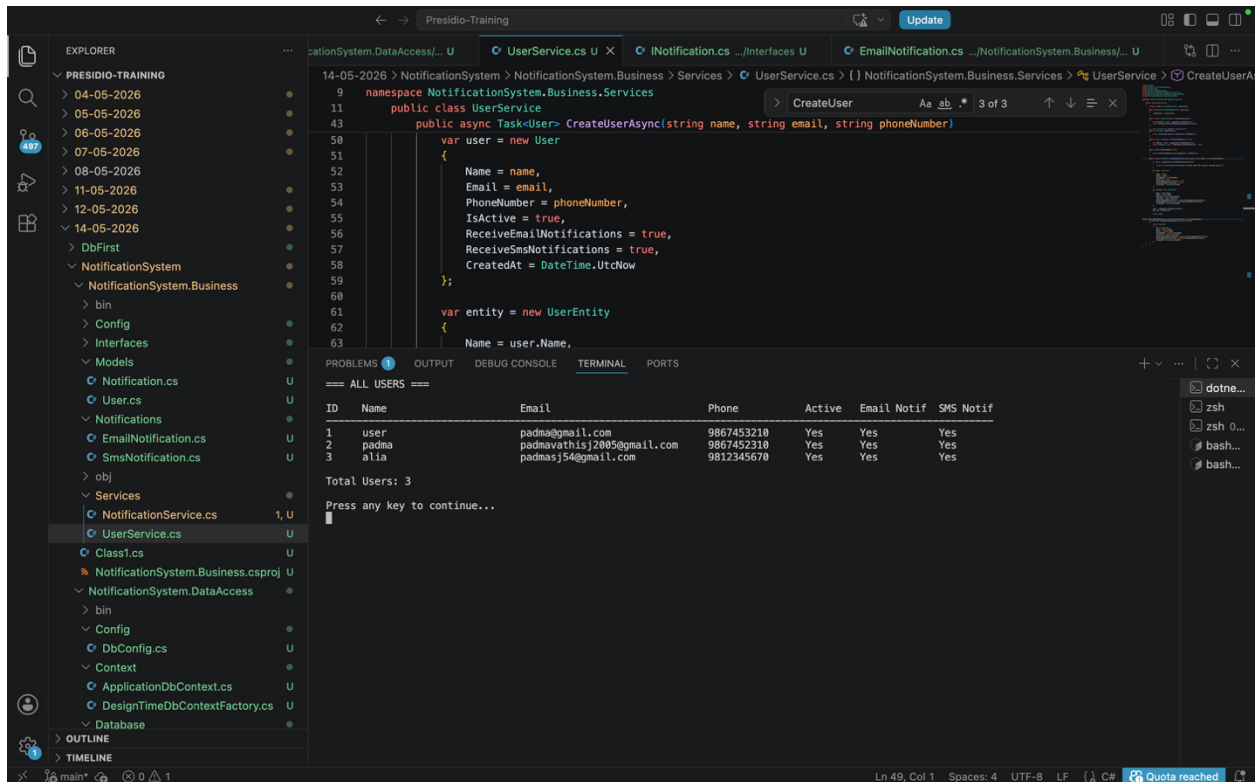
Name: alia
Email: padmasj54@gmail.com
Phone number (10 digits): 9812345670
[DEBUG] CreateUser called for: alia, padmasj54@gmail.com
[DEBUG] Calling repository AddAsync...
[DEBUG] Adding user: alia, padmasj54@gmail.com
[DEBUG] Repository AddAsync completed
[DEBUG] User created with ID: 0

User created successfully with ID: 0

Press any key to continue...

[DEBUG] SaveChangesAsync returned: 1 rows affected
[DEBUG] New user ID: 3

View all users:



```
14-05-2026 > NotificationSystem > NotificationSystem.Business > Services > UserService.cs > {} NotificationSystem.Business.Services > UserService > CreateUserAsync
9 namespace NotificationSystem.Business.Services
11 public class UserService
43 public async Task<User> CreateUserAsync(string name, string email, string phoneNumber)
50 {
51     var user = new User
52     {
53         Name = name,
54         Email = email,
55         PhoneNumber = phoneNumber,
56         IsActive = true,
57         ReceiveEmailNotifications = true,
58         ReceiveSmsNotifications = true,
59         CreatedAt = DateTime.UtcNow
60     };
61     var entity = new UserEntity
62     {
63         Name = user.Name,
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

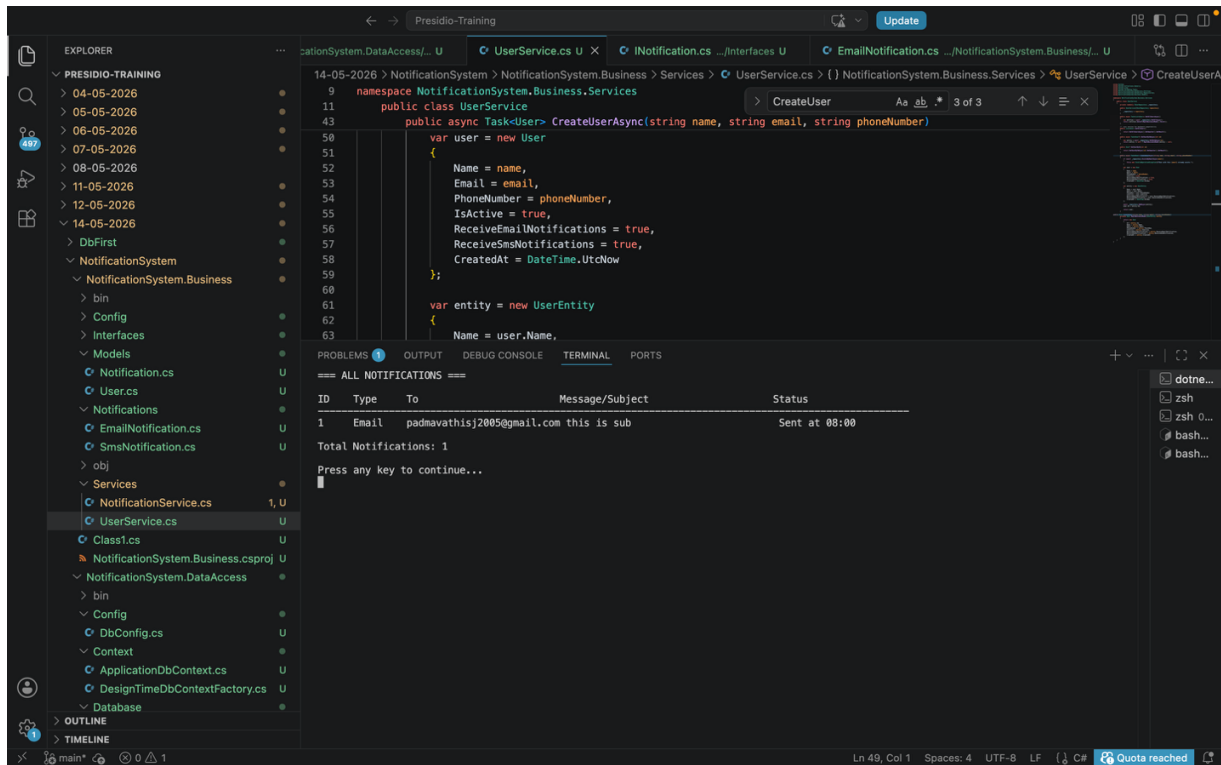
=== ALL USERS ===

ID	Name	Email	Phone	Active	Email Notif	SMS Notif
1	user	padma@gmail.com	9867453210	Yes	Yes	Yes
2	padma	padmavathisj2005@gmail.com	9867452310	Yes	Yes	Yes
3	alia	padmasj54@gmail.com	9812345670	Yes	Yes	Yes

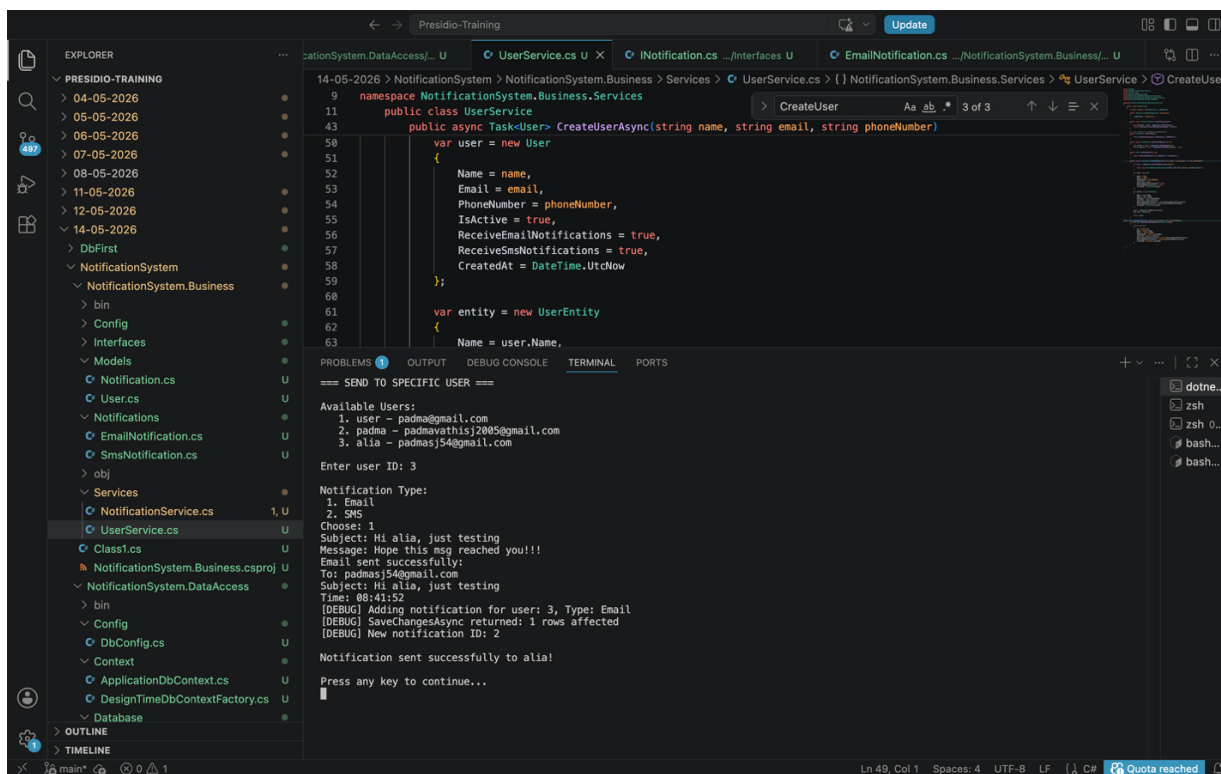
Total Users: 3

Press any key to continue...

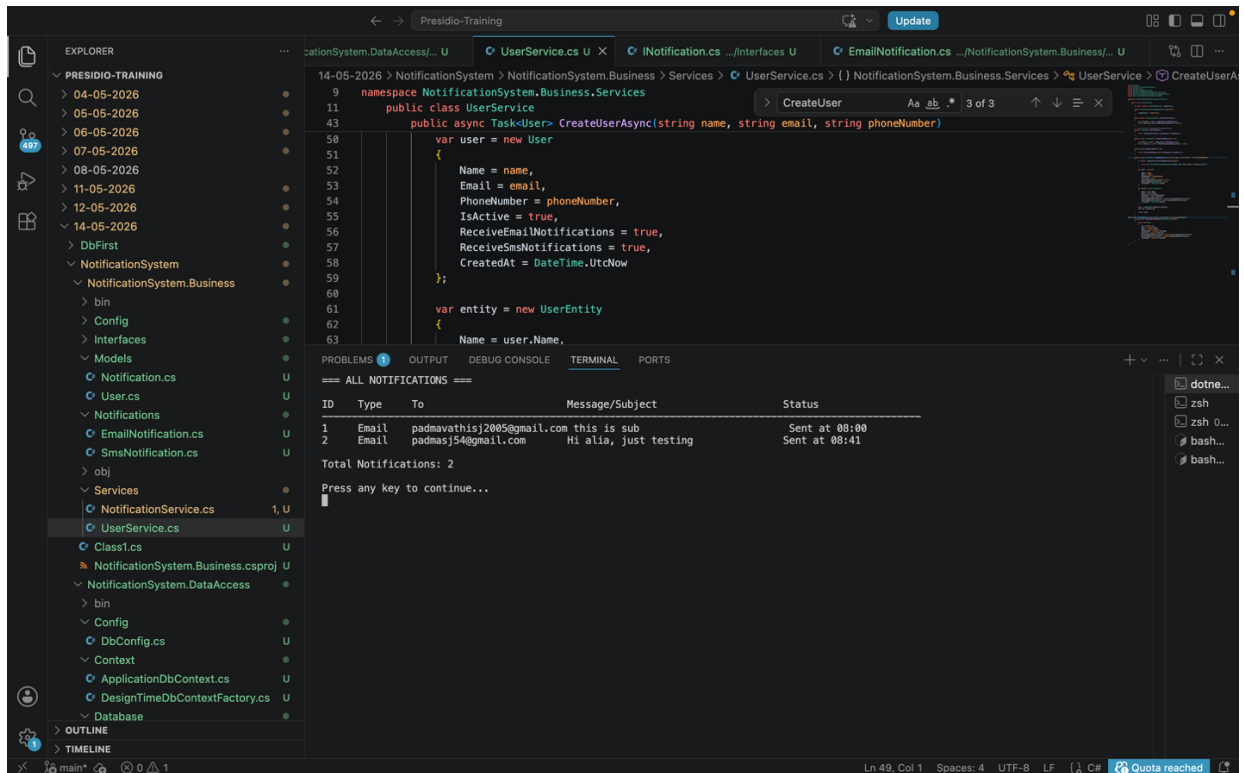
View all sent notifications before sending notification to newly created user(alia):



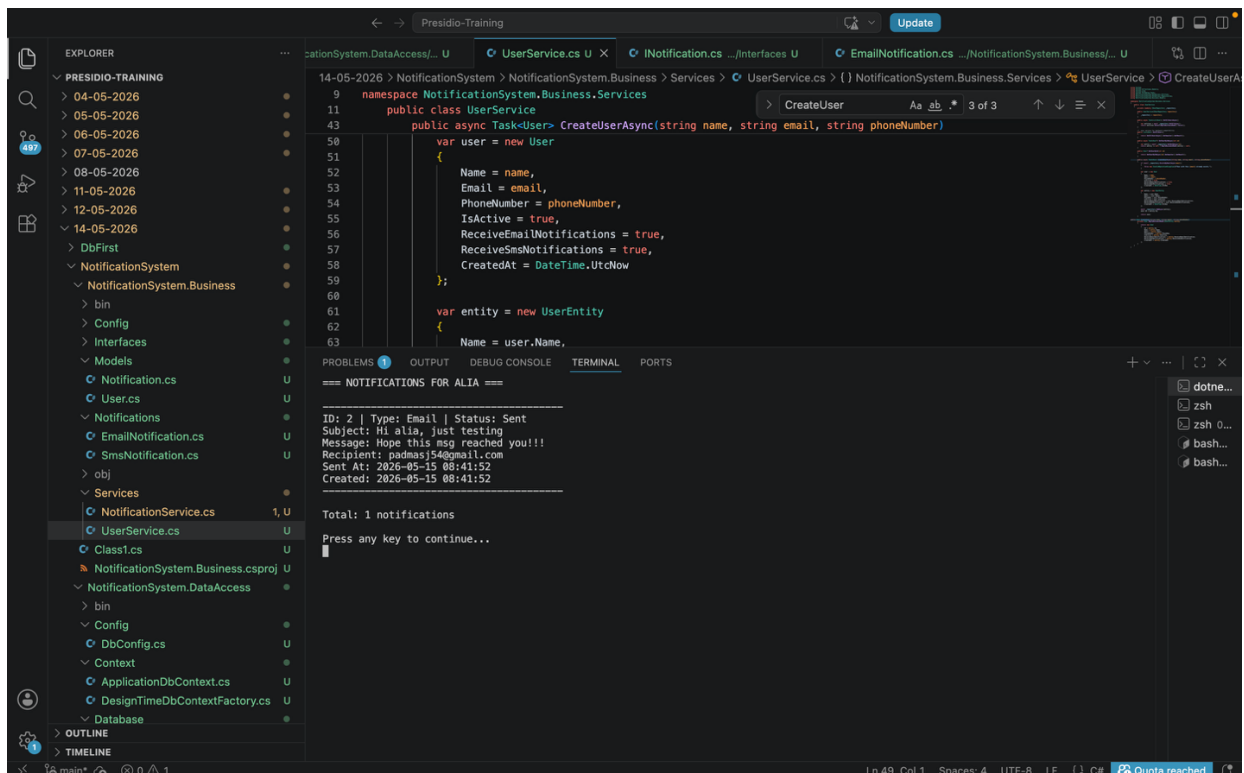
Send notification to specific user(alia):



View all sent notifications after sending notification to newly created user(alia):



View notifications of a specific user:



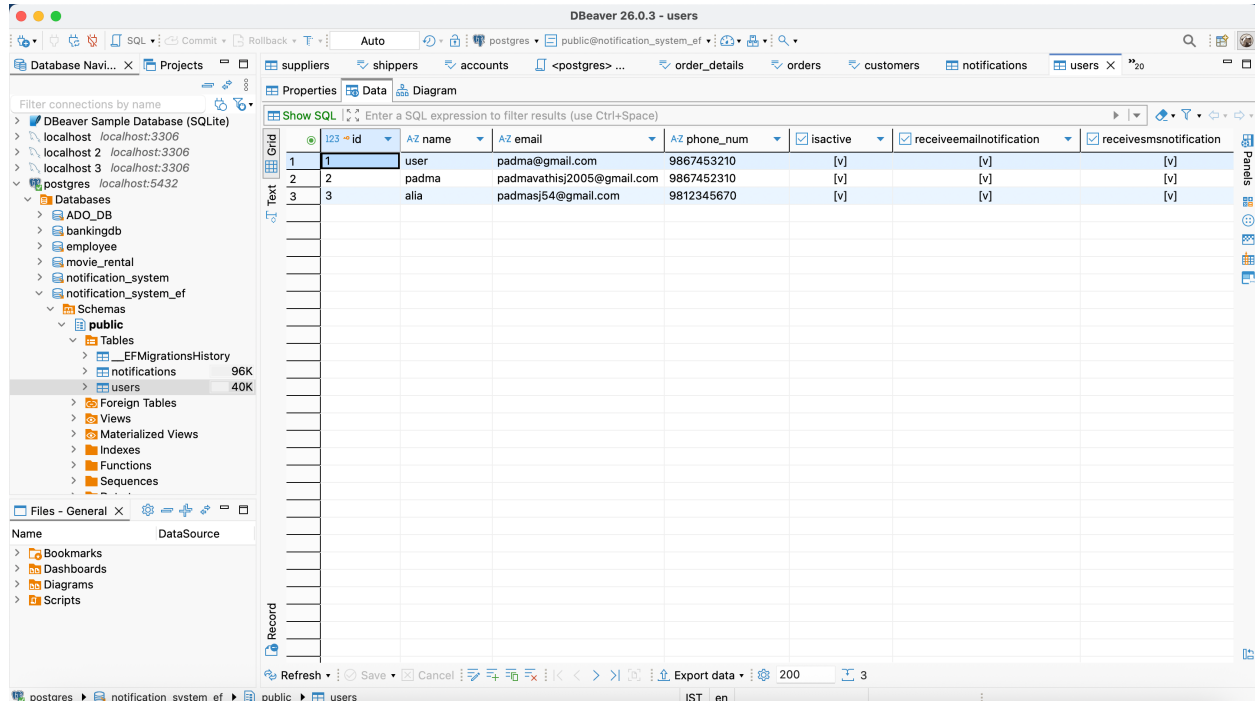
Send notifications to all users:

```
14-05-2026 > NotificationSystem > NotificationSystem.Business > Services > UserService.cs > {} NotificationSystem.Business.Services > UserService > CreateUserA...
9 namespace NotificationSystem.Business.Services
11 public class UserService
...
49
50
51     var user = new User
52     {
53         Name = name,
54         Email = email,
55         PhoneNumber = phoneNumber,
...
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
=== SEND TO ALL USERS ===
Will send to 3 active user(s).
Notification Type:
1. Email
2. SMS
Choose: 1
Subject: this subject is for all users
Message: this msg will tell you some secrets padma, alia, user
Email sent successfully:
To: padma@gmail.com
Subject: this subject is for all users
Time: 08:44:25
[DEBUG] Adding notification for user: 1, Type: Email
[DEBUG] SaveChangesAsync returned: 1 rows affected
[DEBUG] New notification ID: 3
Email sent successfully:
To: padma@gmail.com
Subject: this subject is for all users
Time: 08:44:29
[DEBUG] Adding notification for user: 2, Type: Email
[DEBUG] SaveChangesAsync returned: 1 rows affected
[DEBUG] New notification ID: 4
Email sent successfully:
To: padma@gmail.com
Subject: this subject is for all users
Time: 08:44:33
[DEBUG] Adding notification for user: 3, Type: Email
[DEBUG] SaveChangesAsync returned: 1 rows affected
[DEBUG] New notification ID: 5
Summary ---
Sent: 3
Failed: 0
Total: 3
Press any key to continue...
```

View all sent notifications:

```
14-05-2026 > NotificationSystem > NotificationSystem.Business > Services > UserService.cs > {} NotificationSystem.Business.Services > UserService > CreateUserA...
9 namespace NotificationSystem.Business.Services
11 public class UserService
...
49
50
51     var user = new User
52     {
53         Name = name,
54         Email = email,
55         PhoneNumber = phoneNumber,
...
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
=== ALL NOTIFICATIONS ===
ID Type To Message/Subject Status
1 Email padma@gmail.com this is sub Sent at 08:00
2 Email padma@gmail.com Hi alia, just testing Sent at 08:41
3 Email padma@gmail.com this subject is for all users Sent at 08:44
4 Email padma@gmail.com this subject is for all users Sent at 08:44
5 Email padma@gmail.com this subject is for all users Sent at 08:44
Total Notifications: 5
Press any key to continue...
```

Users table:



DBeaver 26.0.3 - users

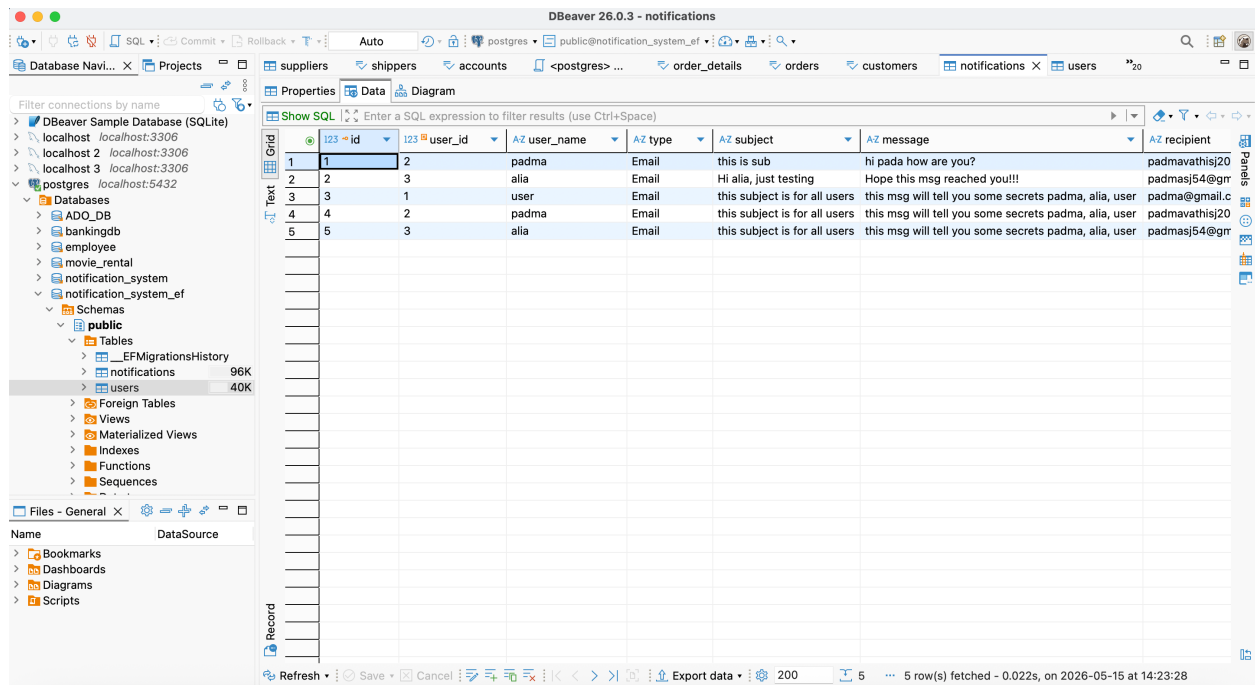
Database Navigator: Filter connections by name. DBeaver Sample Database (SQLite), localhost:3306, postgres:localhost:5432. Databases: ADO_DB, bankingdb, employee, movie_rental, notification_system, notification_system_ef. Schemas: public. Tables: __EFMigrationsHistory, notifications (96K), users (40K). Foreign Tables, Views, Materialized Views, Indexes, Functions, Sequences.

Properties: Data. Show SQL: Enter a SQL expression to filter results (use Ctrl+Space).

	id	name	email	phone_num	isactive	receiveemailnotification	receivesmsnotification
1	1	user	padma@gmail.com	9867453210	[v]	[v]	[v]
2	2	padma	padmavathis2005@gmail.com	9867452310	[v]	[v]	[v]
3	3	alia	padmasj54@gmail.com	9812345670	[v]	[v]	[v]

Refresh Save Cancel Export data 200 3

Notifications table:



DBeaver 26.0.3 - notifications

Database Navigator: Filter connections by name. DBeaver Sample Database (SQLite), localhost:3306, postgres:localhost:5432. Databases: ADO_DB, bankingdb, employee, movie_rental, notification_system, notification_system_ef. Schemas: public. Tables: __EFMigrationsHistory, notifications (96K), users (40K). Foreign Tables, Views, Materialized Views, Indexes, Functions, Sequences.

Properties: Data. Show SQL: Enter a SQL expression to filter results (use Ctrl+Space).

	id	user_id	user_name	type	subject	message	recipient
1	1	2	padma	Email	this is sub	hi pada how are you?	padmavathis20
2	2	3	alia	Email	Hi alia, just testing	Hope this msg reached you!!!	padmasj54@gr
3	3	1	user	Email	this subject is for all users	this msg will tell you some secrets padma, alia, user	padma@gmail.c
4	4	2	padma	Email	this subject is for all users	this msg will tell you some secrets padma, alia, user	padmavathis20
5	5	3	alia	Email	this subject is for all users	this msg will tell you some secrets padma, alia, user	padmasj54@gr

Refresh Save Cancel Export data 200 5 5 row(s) fetched - 0.022s, on 2026-05-15 at 14:23:28

